Tutorial 2: Flow Over a Cylinder

Sajeev
CLL769: Applications of CFD

August 2025

1 Problem Statement & Objective

The objective is to perform 2D numerical simulations of unsteady flow of a Newtonian, incompressible fluid over a cylinder to study vortex shedding. The system consists of a domain of 15×5 m with a cylinder of diameter $D = 1$ m located 5 m from the inlet. Flow regimes studied are Laminar ($Re = 272$) and Turbulent ($Re = 2902$) with an inlet velocity $U = 0.5$ m/s and density $\rho = 1$ kg/m^3.

2 Computational Setup

A velocity inlet of 0.5 m/s and a pressure outlet (0 gauge) were used, alongside no-slip conditions on walls and the cylinder. A pressure-based, transient, planar solver was utilized with the SIMPLE algorithm, QUICK momentum discretization, and 2nd Order Implicit transient formulation. The laminar case utilized the Laminar viscous model, while the turbulent case utilized the Standard $k - \epsilon$ model.

3 Geometry & Meshing

An unstructured mesh with inflation layers near the cylinder was generated. The grid is highly refined near the cylinder walls to capture the boundary layer separation and wake dynamics accurately, while far-field regions are slightly coarser to optimize computational cost.

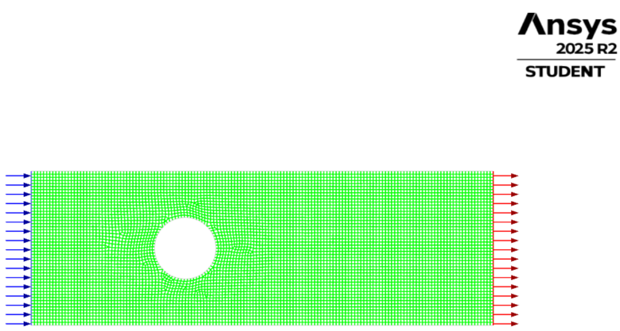

Figure 1: Computational grid around the cylinder.

4 Results & Discussion

4.1 Velocity Contours: Vortex Street

At $Re = 272$, distinct alternating vortices (Von Kármán vortex street) are shed. When switching to the Standard $k - \epsilon$ model ($Re = 2902$), the vortex shedding is heavily damped and the wake is smeared, a known limitation of the standard RANS formulation due to excessive turbulent viscosity production in the wake.

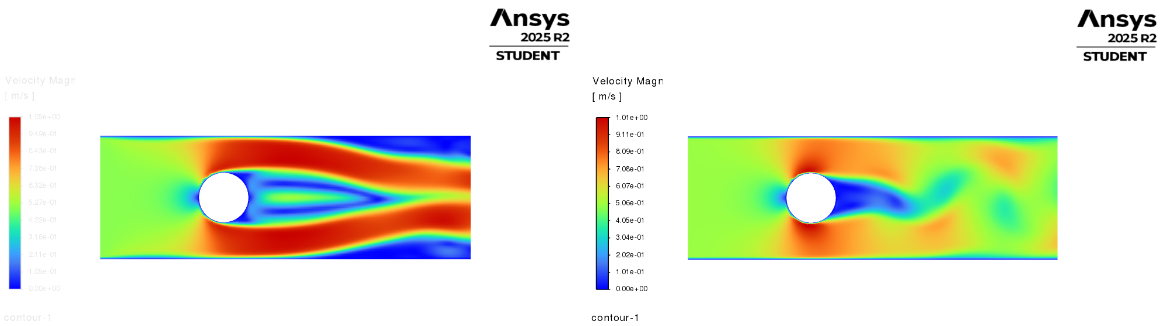

Figure 2: Velocity contours. Left: Laminar ($Re = 272$). Right: Turbulent ($Re = 2902$).

4.2 Frequency Analysis

The y-velocity at a point behind the cylinder was monitored. For the laminar case ($Re = 272$), the oscillation period T from the probe data was approximately 11 s, yielding a shedding frequency $f \approx 0.091$ Hz. This perfectly aligns with the analytical expected Strouhal number $St = fD/U \approx 0.183$ (which predicts $f = 0.183 \times 0.5/1 = 0.0915$ Hz). For the turbulent case, the Standard $k - \epsilon$ model damped out the oscillations entirely, yielding a steady symmetric wake.

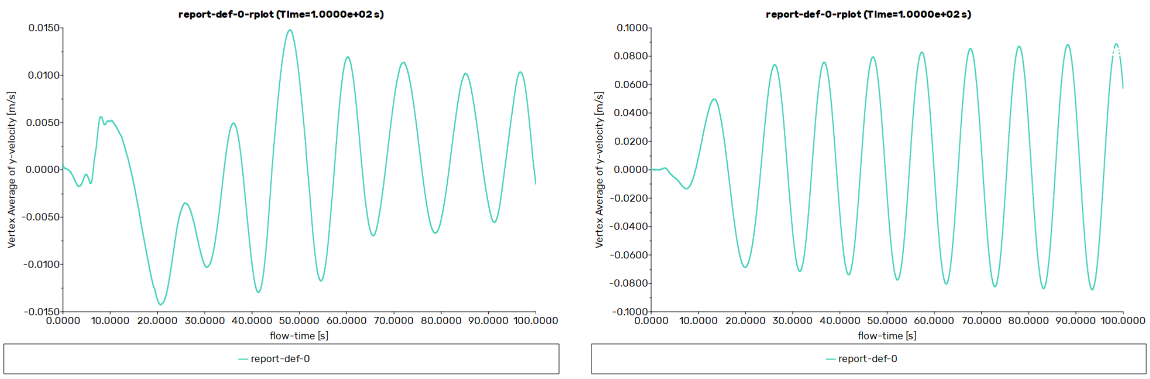

Figure 3: Frequency analysis probe data. Left: Laminar. Right: Turbulent.

4.3 Animation

A solution animation was recorded during the transient simulation to capture the dynamics of the wake formation and the vortex shedding process over time. **Watch Animation:** Click here to view the video on GitHub!